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Claremont McKenna College

Maximizing the MLB Draft: How College Performance and
Leverage Impact Player Signing Bonus and Team Payoff

submitted to
Professor Richard Burdekin

by
Derek Seneker

for
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Abstract

This thesis examines the factors that enable NCAA Division I college position players to maximize their signing bonuses in the MLB draft. I focus on how performance metrics, player characteristics, and program-related attributes shape bonus outcomes relative to draft slot values. Using data from the top five rounds of MLB drafts from 2012 to 2022, this research provides insights into optimal valuation strategies for draft prospects. Additionally, it explores whether higher signing bonuses correlate with a player's likelihood of eventually making their MLB debut, assessing the potential return on investment for MLB teams. By filling a gap in existing literature, this study presents a framework that benefits players in maximizing their bonuses and teams in making strategic draft investments.

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1. Introduction

The Major League Baseball (MLB) draft plays a significant role in shaping the careers of aspiring professional baseball players and is a key moment for MLB teams looking to build championship-level rosters. It marks the start of professional careers for players while offering teams an opportunity to acquire talent within financial constraints. The draft operates as a structured process to distribute young prospects fairly, with the selection order determined by a combination of a lottery system, team performance, and revenue-sharing status. This system promotes competitive balance by giving lower-performing teams priority access to the best prospects. At the same time, the draft's bonus pool and slot value system regulate spending and ensure parity, though they often limit players' financial opportunities. For most draftees, signing bonuses are a critical financial milestone, representing the bulk of their income during their early careers in the minor leagues, where salaries are notoriously low. Additionally, the negotiation dynamics vary significantly between high school and college players, with high school athletes often holding greater leverage due to alternative options like college commitments. The emergence of Name, Image, and Likeness (NIL) deals has further reshaped the decision-making process for players, adding complexity to their financial considerations. To fully understand the MLB draft's impact, it is essential to examine its structure, the financial systems governing it, and the negotiation strategies that define player-team relationships.

Draft Structure and Competitive Balance

Every year, teams select ballplayers from high schools and colleges via a structured drafting process. The draft is designed to distribute young talent, with the selection order determined by a combination of a lottery system, regular season and postseason standings, and

revenue-sharing status, ensuring that lower-performing teams have the opportunity to select the best prospects. This system is designed to promote competitive balance across the league, allowing all teams a fair chance to build and maintain a competitive roster.

In contrast to the restrictive yet fair regulations of the draft, MLB's free agency system allows players and teams to negotiate contracts in an open market without restrictions. High-revenue teams dominate free agency, offering substantial long-term deals that smaller-market teams often cannot afford, exacerbating competitive imbalances. The draft's bonus pool and slot value system were introduced to curb excessive spending and create fairness, ensuring all teams have an equal chance to sign top prospects. While this system effectively controls costs and promotes balance, it also limits drafted players' financial opportunities, as their bonuses are tied to a system that has not evolved alongside the league's financial growth. For many players, slot values act as a ceiling on their earning potential, limiting their ability to negotiate higher bonuses. This limitation is especially significant given the low wages typically earned in the minor leagues, where signing bonuses often make up the bulk of a player's income during the early stages of their career. For many players, the signing bonus may end up being the majority or entirety of their career earnings, as only 5% of those drafted ever make it to Major League Baseball and secure higher-paying contracts. While all drafted players are selected by MLB teams, most spend their careers in the minor leagues, where salaries are significantly lower.

For players, the draft is the culmination of years of dedication and the final step toward realizing their dream of playing professional baseball. It also marks a pivotal financial milestone, providing their first professional paycheck and establishing a foundation for future financial security. For MLB teams, the draft presents a unique opportunity to invest in future All-Stars, MVPs, and Hall of Famers at a fraction of the cost of the free-agent market. Unlike free agency,

where wealthier teams can outspend others for elite talent, the draft ensures every team has the chance to select high-potential players who can make a significant impact.

Eligibility for the MLB draft depends on a player's age and level of education. High school players can declare for the draft immediately after graduating. College players, however, must wait until they are at least 21 years old or have completed their junior year to become eligible. Leading up to an eligible draft year, most players work with agents to handle negotiations. Agents play a critical role in advocating for players by leveraging performance data, scouting reports, and projected potential to negotiate the best possible terms. For high school players, agents often emphasize their option of choosing to attend college if terms are unfavorable. College players rely on agents to highlight their performance against college talent and readiness to transition to professional play.

Bonus Pools and Slot Values

The MLB draft operates within a structured system that allocates signing bonuses to newly-drafted players, aiming to regulate spending and promote competitive balance. The bonus pool system, introduced in 2012 as part of a new collective bargaining agreement, marked a significant change in how teams approached the draft. It established a more structured approach to managing signing bonuses and allocating resources.

Each year, the league's total bonus pool is determined based on factors such as overall revenue and the terms of the collective bargaining agreement. In 2024, for instance, the total draft bonus pool was \$330,742,700, with individual team allocations ranging from \$18,334,000 for the Cleveland Guardians to \$5,914,700 for the Houston Astros, depending on the number and

placement of their draft picks. The pool is distributed proportionally, with teams holding earlier picks or additional selections receiving larger shares.

Each draft pick is assigned a recommended signing bonus, known as the "slot value," which serves as a guideline for compensation. For example, the first overall pick in the 2024 draft, Travis Bazzana, had a slot value of \$10,570,600 but signed for \$8,950,000, demonstrating how actual bonuses can deviate from these benchmarks. For picks beyond the 10th round, only the portion of a signing bonus exceeding \$150,000 counts toward the pool, giving teams the flexibility to pursue later-round talent with minimal impact on their budget. However, exceeding the bonus pool triggers significant penalties, ranging from a 75% tax on minor overages to the forfeiture of future draft picks for spending more than 15% above the cap. These rules require teams to carefully balance their financial strategies with talent acquisition.

A player's signing bonus is often determined by two main factors: performance and leverage. Strong performance metrics and standout skills can make a prospect highly appealing to teams, potentially allowing for negotiations above the assigned slot value. However, leverage, the ability for a player to walk away or pursue alternative paths, plays a crucial role as well.

Negotiation Dynamics

High school players often have a unique leverage advantage in draft negotiations, enabling them to negotiate signing bonuses above their slot value more frequently than college players. Most high school prospects have already been recruited by top college programs and committed to playing for a specific school before entering the draft. This commitment gives them a strong fallback option if they receive an unfavorable offer. By choosing to honor their college plans, high school athletes can continue their education, train in elite facilities, and

re-enter the draft at 21 with improved skills and a potentially higher value provided they develop as expected and avoid significant injuries. The college acknowledges that the player might delay or forgo enrollment to sign professionally, and the MLB team knows the player might reject their offer to pursue college.

College players do not have the same negotiation power as high school athletes, especially college players who have no remaining NCAA eligibility. Each additional year they stay on campus after becoming draft-eligible reduces their leverage because they have fewer alternatives to the draft. College players are perceived as more pro-ready due to their experience playing high-level competition, but with little room for development due to their age. Given the bonus pool constraint and teams often signing high school players above slot bonuses, college prospects typically see a smaller bonus than their predicted slot value.

The recent introduction of Name, Image, and Likeness (NIL) deals in college sports has added another layer to this dynamic, offering student-athletes an additional income stream. NIL deals allow college athletes to earn money through sponsorships, endorsements, and other opportunities, fundamentally altering the landscape of collegiate sports. For high school athletes considering college, NIL earnings make the prospect of attending more appealing, providing a chance to earn money while developing their skills and gaining exposure. Similarly, for college athletes, NIL opportunities provide a financial incentive to remain in school longer. However, despite the growing significance of NIL deals, the financial rewards of an MLB signing bonus often far outweigh the earnings from NIL sponsorships, particularly for players securing substantial bonuses. This highlights the enduring importance of the draft in shaping an athlete's financial future and emphasizes the strategic considerations athletes face when deciding between professional baseball and collegiate opportunities.

Key Findings

Although much research has explored draft positioning and signing bonuses, few studies have examined the discrepancies between slot values and actual bonuses. Bonuses exceeding slot values may indicate heightened confidence in a player's potential, while below-slot bonuses could reflect more conservative investments by teams. This study examines the possible factors explaining this, and whether these discrepancies predict future MLB success for college players, addressing the overlooked connection between signing bonuses and career outcomes.

By focusing on performance metrics, player characteristics, and collegiate program status, this research contributes to existing literature. Specifically, it analyzes NCAA Division I college position players selected in the top five rounds of the MLB draft. Previous studies, such as Burger and Walters (2009), highlight the general success rates of college players versus high school players but do not investigate the drivers of bonus discrepancies. Hubley (2012) critiques the inefficiencies of the slot system but does not focus on its specific effects on college athletes. Similarly, while Spurr (2000) and Winfree and Molitor (2007) examine negotiation dynamics and fallback options for high school players, they overlook the unique challenges faced by college players with limited leverage. This study fills these gaps by investigating how bonuses deviate from slot values and whether above-slot bonuses correlate with a higher likelihood of making an MLB debut. It aims to provide actionable insights for college position players to maximize their draft outcomes and for MLB teams to enhance their investment strategies.

Key findings of this study reveal that performance and leverage significantly influence financial outcomes in the MLB draft, though their ability to predict long-term success is limited. Players with strong power-hitting metrics, particularly home runs, tend to secure higher bonuses. Eligibility also impacts negotiations, as advancing in grade level often results in reduced signing

bonuses. However, financial outcomes, such as signing bonuses, show limited predictive power for MLB success. Minimal differences in MLB debut rates between players signing above and below slot value suggest that factors like minor league performance and organizational development play a more substantial role in career trajectories. These results emphasize the importance of age, leverage, and on-field performance in shaping financial outcomes while underscoring the need to explore broader variables to better predict long-term success.

The following sections analyze previous literature, describe the datasets used, outline the methodology, and present findings and implications. A review of existing studies highlights the gaps this research addresses. The dataset and analytical models are detailed, followed by key findings that illustrate significant trends and relationships. Finally, the discussion offers practical implications for players and teams, and the paper concludes with a summary of findings and suggestions for future research directions.

2. Literature Review

Existing research has examined the MLB draft's structure, its slot and bonus allocation systems, and the factors predicting MLB success. However, a notable gap remains in understanding the specific factors driving discrepancies between signing bonuses and slot values, particularly for college position players. This study aims to address this gap by analyzing how performance metrics and player characteristics influence these discrepancies and potentially predict MLB outcomes.

Epstein (2024) critiques the MLB's bonus pool system, noting that while it is designed to promote competitive balance, it often restricts college players' ability to secure fair bonuses. The system's emphasis on potential disproportionately favors high school players, who are perceived to have greater long-term growth potential. As a result, college players, despite their more refined skills and experience, frequently face financial disadvantages in the draft.

Comparing college and high school players' likelihood of reaching the MLB, Burger and Walters (2009) find that college players have a 57% higher likelihood of success compared to high school players, highlighting their readiness for professional play. College players' maturity and experience often make them more reliable investments for MLB teams. However, systemic biases that prioritize the speculative potential of high school players often result in lower bonuses for college prospects.

Hubley (2012) expands on these disparities, noting that "college players benefit from up to four additional years of coaching and training" (p. 4) and extended scouting exposure, which provides teams with higher-quality information compared to high school prospects. As Hubley explains, "college hitters, within the first few rounds, provide the most return (in terms of value added) than any other type of draft pick" (p. 15). Despite this, the draft system continues to

prioritize high school players' speculative potential, often undervaluing the proven readiness and immediate contributions of collegiate athletes, leading to inefficiencies in resource allocation.

Spurr (2000) underscores the reliability of college players in the draft, reporting that “almost 72% of the players selected in the June draft were college players” (p. 75). This success rate demonstrates the stability and readiness college players offer, making them more dependable investments for teams. However, despite their proven performance, high school players often receive higher bonuses due to the belief in their greater developmental potential and long-term growth.

Winfree and Molitor (2007) analyze negotiation dynamics, highlighting how high school players leverage their fallback option of attending college to secure better bonuses. As they note, “High school players who are drafted must decide between entering college and entering the minor leagues. If they play baseball professionally right after high school, they lose eligibility to play college baseball. However, many players feel that their best chance to play major league baseball is to enter the professional ranks right after high school. Also, extremely attractive signing bonuses lure athletes to turn professional as early as possible” (p. 379). While college provides the opportunity to earn a degree and increase non-baseball earnings, “the risk for injury in college may give players an incentive to play professional baseball sooner” (p. 379). Conversely, college players who exhaust their eligibility lack this leverage, often forcing them to accept bonuses at or below slot values.

Ahmadi et al. (2022) explores the impact of spatial proximity on MLB draft decisions, revealing that players living closer to scouting directors or team locations are more likely to be drafted, even after controlling for skill. This geographic closeness influences financial outcomes, as players living in the same state as the scouting director receive 23.85% higher signing bonuses

on average, highlighting the role of proximity in negotiations. However, the study also identifies significant inefficiencies associated with proximity bias, as players drafted based on geographic closeness are 38% less likely to make it to the MLB compared to those selected without this bias. Propinquity bias is particularly pronounced in later rounds of the draft, where scouting directors have greater decision-making autonomy and are more likely to select players from nearby locations.

In conclusion, this study addresses a critical gap in understanding the factors driving discrepancies between signing bonuses and slot values for college position players in the MLB draft. While existing research highlights the draft's structure, its biases, and the comparative advantages of college players, it also underscores systemic inefficiencies that prioritize high school prospects' potential over the proven readiness of collegiate athletes. By analyzing performance metrics, player characteristics, and negotiation dynamics, this study provides valuable insights into how these factors influence financial outcomes and the likelihood of MLB success. These findings can help college players better navigate the draft process and inform teams seeking to optimize their draft strategies, ensuring more equitable and efficient resource allocation in future drafts.

3. Data

Data Sources and Selection Criteria

The data used in this study are drawn from multiple sources, including Spotrac (<https://www.spotrac.com>) for MLB draft data for the years 2012–2022 and The Baseball Cube (<https://www.thebaseballcube.com>) for player statistics. Only NCAA Division I college position players selected in the top five rounds of the draft were included to ensure a comparable sample of high-caliber prospects. To maintain data quality and relevance, several steps of data cleaning were applied. Only statistics from a player's final year of college were included to ensure consistency and relevance to their draft-year performance. A minimum signing bonus of \$100,000 was established to focus on players with larger financial stakes, and players who did not come to an agreement with a team on their signing bonuses were excluded.

To further refine the dataset, players were required to have played a minimum of 30 games to ensure their statistics were representative of meaningful performance. Additionally, data from non-Division I players and the COVID-affected 2020 season were omitted to avoid discrepancies caused by performance anomalies and irregular draft decisions. Finally, *DIFF*, representing the percentage difference between a player's actual signing bonus and their assigned slot value, was used to examine signing bonus discrepancies. *DIFF* is calculated as:

$$DIFF = \frac{Actual\ Signing\ Bonus - Slot\ Value}{Slot\ Value} \times 100$$

Players with *DIFF* values falling outside the 25th and 75th percentiles were excluded to remove extreme outliers, such as unusually high or low bonuses, which often result from unique circumstances like injury concerns or extraordinary negotiation leverage. This step ensures the analysis focuses on typical trends in signing bonuses while minimizing distortions caused by

anomalies. Remaining missing values were checked for consistency across sources, requiring minimal adjustments due to comprehensive initial filtering.

Variables and Summary Statistics

To investigate the factors influencing the percentage change between MLB draft signing bonuses and slot values (*DIFF*), a comprehensive dataset was assembled. This dataset includes variables categorized into three main groups: player demographics and draft details, collegiate performance metrics, and college program characteristics. Each variable was selected to capture relevant aspects of a player's background, performance, and exposure to competitive environments. To evaluate whether a higher signing bonus translates into a tangible return for a team, the analysis includes an MLB debut variable. Below is a detailed list of the variables used in the analysis.

Player Demographics

DIFF: Percentage change from signing bonus to slot value (dependent variable).

AGE: Player's age (years).

GRADE: Player's academic standing (1 = freshman, 2 = sophomore, 3 = junior, 4 = senior).

CATCHER: Catcher (0 = no, 1 = yes).

FIRST: First baseman (0 = no, 1 = yes).

SECOND: Second baseman (0 = no, 1 = yes).

THIRD: Third baseman (0 = no, 1 = yes).

SHORT: Shortstop (0 = no, 1 = yes).

OUTFIELD: Outfielder (0 = no, 1 = yes).

Collegiate Performance Metrics

R: Runs scored.

SING: Singles hit.

DOUB: Doubles hit.

TRIP: Triples hit.

HR: Home runs.

RBI: Runs batted in.

SB: Stolen bases.

AVG: Batting average.

OPS: On-base plus slugging percentage.

BABIP: Batting average on balls in play.

SOBB: Strikeout-to-walk ratio.

RC: Runs created.

College Program Characteristics

NCAATOURN: Participation in the NCAA Tournament (0 = no, 1 = yes).

CWS: Participation in the College World Series (0 = no, 1 = yes).

PFIVE: Affiliation with a Power Five conference (0 = non-Power Five, 1 = Power Five).

Valuable Return

DEBUT: Indicates whether the player has made their MLB debut (0 = no, 1 = yes).

Table 1: Summary Statistics

Variable	Obs	Mean	Std. dev.	Min	Max
DIFF	385	-.0746739	.1110458	-.3995797	.2385321
AGE	385	20.90909	.4559165	19	22
GRADE	385	2.812987	.4636109	1	4
NCAATOURN	385	.6051948	.4894449	0	1
CWS	385	.1662338	.3727748	0	1
PFIVE	385	.6779221	.4678807	0	1
CATCHER	385	.1428571	.3503824	0	1
FIRST	385	.0883117	.2841169	0	1
SECOND	385	.1116883	.3153925	0	1
THIRD	385	.1116883	.3153925	0	1
SHORT	385	.1584416	.3656297	0	1
OUTFIELD	385	.387013	.4877005	0	1
R	385	51.74026	11.68243	15	85
SING	385	46.75325	11.08814	17	85
DOUB	385	15.05714	4.452511	5	30
TRIP	385	2.158442	1.962762	0	11
HR	385	10.56104	5.784952	0	32
RBI	385	48.56364	13.92621	9	94
SB	385	9.818182	8.07763	0	44
AVG	385	.3334753	.0379391	.212	.471
OBP	385	.4307091	.0451069	.314	.619
SLG	385	.5629169	.0985365	.328	.928
OPS	385	.993626	.132322	.642	1.547
BABIP	385	.3604727	.0436083	.222	.516
BB	385	12.71268	4.186192	3.47	28.57
SO	385	14.32777	4.812221	2.46	28.47
SOBB	385	1.236	.5606782	.19	3.64
RC	385	53.43495	14.11099	19.64021	105.5733

The dataset consists of 385 NCAA Division I position players drafted in the top five rounds of the MLB draft between 2012 and 2022. This cohort represents high-caliber prospects, selected for their standout collegiate performances and potential for professional success. The primary variable of interest, *DIFF*, measures the percentage difference between a player's signing bonus and the slot value assigned to their draft position. On average, *DIFF* is -0.0747, indicating that most players signed slightly below their slot value. The standard deviation of 0.1110 shows moderate variability, with values ranging from -0.3996 to 0.2385, capturing a wide spectrum of negotiation outcomes.

Player ages average 20.91 years, with a range of 19 to 22 and a standard deviation of 0.456. This aligns with the draft eligibility window for collegiate athletes, who are typically drafted after their sophomore or junior year. *GRADE*, reflecting collegiate experience (1 = freshman, 4 = senior), averages 2.81, indicating that juniors are the dominant group. Juniors often have the most leverage in negotiations due to their combination of experience and remaining eligibility, while freshmen and seniors are less represented due to limited exposure or reduced bargaining power.

Participation in high-profile collegiate events significantly enhances players' visibility. A majority (60.5%) of players participated in the NCAA Tournament, highlighting the importance of postseason exposure in draft evaluations. Additionally, 16.6% of players competed in the prestigious College World Series (CWS), further increasing their visibility on a national stage. Furthermore, 67.8% of players came from *PFIVE* conferences, which include the ACC, Big Ten, Big 12, Pac-12, and SEC. These conferences are distinguished by their superior athletic programs, advanced coaching, and access to top-tier facilities, which better prepare players for the challenges of professional baseball. Players in *PFIVE* conferences also face high levels of

competition and receive greater attention from MLB scouts, making their performance metrics more reliable indicators of potential MLB success.

Offensive performance metrics demonstrate the players' capabilities at the plate. *OPS* (on-base plus slugging percentage), a composite measure of offensive productivity, averages 0.994, with values ranging from 0.642 to 1.547. This range captures both solid contributors and elite hitters. The batting average (*AVG*) averages 0.333, with values spanning from 0.212 to 0.471, indicating consistent and effective hitting across the sample. *Runs Created (RC)*, a metric estimating total offensive contributions, averages 53.43, with a maximum of 105.57, showcasing the impact of standout performers.

Additional offensive statistics illustrate the diversity of skills among the players. *Home Runs (HR)* average 10.56, with a maximum of 32, representing a mix of power hitters and those with other strengths. *Stolen Bases (SB)* average 9.82, with a maximum of 44, reflecting the contributions of speed-oriented players. Players also average 15.06 *Doubles (DOUB)* and 2.16 *Triples (TRIP)*, indicating well-rounded offensive outputs.

This dataset provides a comprehensive overview of collegiate baseball players drafted into the MLB, highlighting how demographic characteristics, competitive exposure, and performance metrics influence signing bonuses and draft negotiation outcomes.

4. Econometric Model

The regression model in this study answers the factors explaining the difference (*DIFF*) between MLB draft signing bonuses and slot values. In this model, i represents an individual player, capturing player-specific values for each variable in the regression.

$$DIFF_i = \beta_0 + \beta_1(AGE_i) + \beta_2(NCAATOURN_i) + \beta_3(PFIVE_i) + \beta_4(HR_i) + \beta_5(OPS_i) + \beta_6(BABIP_i) + \beta_7(SOBB_i) + \varepsilon_i$$

To improve the accuracy and interpretability of the regression analysis, I removed unnecessary variables that did not contribute significantly to the model or overlapped with other predictors. This process minimized multicollinearity and ensured the inclusion of only the most relevant predictors, such as demographic factors, performance metrics, and exposure-related variables. For example, metrics like *AVG* and *SLG* were excluded as they were strongly correlated with *OPS* and other performance measures, potentially distorting coefficient estimates.

Additionally, all continuous independent variables were measured deviations from their means. This adjustment allowed for a more meaningful interpretation of the regression constant (β_0), ensuring it equaled the mean of *DIFF* (-0.0747). Centering the variables also reduced multicollinearity among predictors, particularly for performance metrics like *BABIP*, *OPS*, and *HR*, enabling the model to more clearly isolate the effects of each variable. This streamlined and adjusted approach provided a robust foundation for analyzing the factors influencing signing bonus deviations in the MLB draft.

The analysis faced several econometric challenges that highlight the complexities of modeling signing bonus discrepancies in the MLB draft. One significant issue was the consistently low R-squared values, which ranged from 0.0137 to 0.0678. These values indicate that the included variables explain only a small portion of the variation in *DIFF*. This limitation

suggests the presence of unobserved factors, such as team-specific financial strategies, subjective scouting evaluations, or broader market dynamics, that likely play a substantial role in determining signing bonuses relative to slot values. While a low R-squared does not invalidate the model, it underscores the need to consider additional variables or external influences in future research.

Omitted variable bias also posed a potential challenge. Factors such as team financial flexibility, scouting preferences, or the presence of high-profile agents were not included in the dataset but are likely to significantly influence signing bonuses. The exclusion of these factors may have introduced bias into the coefficient estimates, reducing the model's accuracy and explanatory power. This limitation highlights the scope of the dataset and the inherent difficulty in capturing all relevant influences on *DIFF*.

Multicollinearity was another issue, particularly among performance metrics like *OPS*, *BABIP*, and *AVG*. These metrics often measure overlapping aspects of a player's offensive capabilities, which can distort coefficient estimates and make it difficult to isolate the independent effects of each variable. To address this, variable selection emphasized distinct and complementary information. For example, *HR* and *BABIP* were prioritized over broader metrics like *AVG*. Additionally, centering the variables by subtracting their means helped reduce correlations among predictors, mitigating the impact of multicollinearity.

Variability in signing bonuses across players also introduced the potential for heteroskedasticity, which violates the assumptions of ordinary least squares regression. Factors such as draft position, performance metrics, and team-specific considerations may contribute to non-constant variance in the residuals. To account for this, robust standard errors were employed, ensuring that coefficient significance tests remained reliable despite potential heteroskedasticity.

These econometric considerations underscore the challenges of modeling *DIFF* in the MLB draft. While steps were taken to address multicollinearity, heteroskedasticity, and variable selection, the analysis remains constrained by the scope of the available data and the complexity of unobserved factors. Despite these limitations, the model provides valuable insights into the relationships between player characteristics, performance metrics, and signing bonus discrepancies, which are further explored in the subsequent analysis.

5. Results

Table 2: Demographic and Program Factors Influencing Signing Bonus Deviations

Source	SS	df	MS	Number of obs	=	385
Model	.089907519	3	.029969173	F(3, 381)	=	2.46
Residual	4.64525804	381	.012192278	Prob > F	=	0.0626
				R-squared	=	0.0190
				Adj R-squared	=	0.0113
Total	4.73516556	384	.01233116	Root MSE	=	.11042

DIFF	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
GRADE_c	-.0293053	.0121648	-2.41	0.016	-.0532239	-.0053867
NCAATOURN_c	.0056132	.0122049	0.46	0.646	-.0183842	.0296107
PFIVE_c	.0130985	.0127597	1.03	0.305	-.0119898	.0381868
_cons	-.0746739	.0056275	-13.27	0.000	-.0857386	-.0636091

Table 3: Performance Metrics Impact on Signing Bonus Deviations

Source	SS	df	MS	Number of obs	=	385
Model	.268877715	4	.067219429	F(4, 380)	=	5.72
Residual	4.46628785	380	.011753389	Prob > F	=	0.0002
				R-squared	=	0.0568
				Adj R-squared	=	0.0469
Total	4.73516556	384	.01233116	Root MSE	=	.10841

DIFF	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
HR_c	.0064007	.002411	2.65	0.008	.0016601	.0111413
OPS_c	-.200841	.1269373	-1.58	0.114	-.4504285	.0487466
BABIP_c	.1054541	.248014	0.43	0.671	-.3821975	.5931057
SOBB_c	.0093534	.0138247	0.68	0.499	-.017829	.0365358
_cons	-.0746739	.0055252	-13.52	0.000	-.0855378	-.06381

Table 4: Combined Effects on Signing Bonus Deviations

Source	SS	df	MS	Number of obs	=	385
Model	.315070558	7	.04501008	F(7, 377)	=	3.84
Residual	4.420095	377	.01172439	Prob > F	=	0.0005
				R-squared	=	0.0665
				Adj R-squared	=	0.0492
Total	4.73516556	384	.01233116	Root MSE	=	.10828

DIFF	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
GRADE_c	-.0232429	.0120439	-1.93	0.054	-.0469246	.0004388
NCAATOURN_c	-.0036978	.0122866	-0.30	0.764	-.0278566	.0204611
PFIVE_c	.0051356	.0128065	0.40	0.689	-.0200455	.0303166
HR_c	.0063523	.0024946	2.55	0.011	.0014473	.0112573
OPS_c	-.2046446	.1311515	-1.56	0.120	-.4625248	.0532356
BABIP_c	.1141831	.2501333	0.46	0.648	-.3776482	.6060143
SOBB_c	.0066248	.0139523	0.47	0.635	-.0208093	.0340588
_cons	-.0746739	.0055184	-13.53	0.000	-.0855246	-.0638232

Three regression models were conducted to analyze the relationship between *DIFF* and various factors. The first regression examined demographic and collegiate experience variables (*GRADE*, *NCAATOURN*, and *PFIVE*). The second regression incorporated performance metrics (*HR*, *OPS*, *BABIP*, and *SOBB*) to assess their influence. Finally, the third regression combined demographic, experience, and performance variables to provide a comprehensive understanding of the factors shaping signing bonus deviations. Each model's results shed light on distinct aspects of the draft process, including negotiation dynamics, player valuation, and team preferences.

The first regression focused on *GRADE*, *NCAATOURN*, and *PFIVE*, exploring how collegiate experience and conference affiliation impact *DIFF*. The model's R-squared value was

0.0190, indicating that these variables explain only 1.90% of the variation in *DIFF*. Although the explanatory power of the model is limited, several key findings emerged.

The variable *GRADE* showed a significant negative relationship with *DIFF* (coefficient: -0.0293, $p = 0.016$). This result indicates that for each one-unit increase in *GRADE* (e.g., moving from sophomore to junior or junior to senior year), *DIFF* decreases by approximately 2.93%. For the average signing bonus in the dataset, which is \$1,239,598.28, this translates to a reduction of \$35,948. In practical terms, players in higher *GRADE* levels, such as seniors, tend to receive bonuses closer to or below their slot values. This trend reflects the leverage advantage of younger players, particularly juniors, whose eligibility status allows for greater bargaining power. Seniors, by contrast, face reduced leverage and are often perceived as having lower developmental upside, which limits their ability to negotiate higher bonuses.

In contrast, *NCAATOURN* and *PFIVE* did not show statistically significant effects on *DIFF*. These results suggest that while players' participation in prestigious tournaments or conferences may enhance their visibility, these factors alone are insufficient to drive significant deviations in signing bonuses.

The low explanatory power of this model highlights the need to incorporate additional variables, particularly performance metrics, to better understand the factors influencing *DIFF*. While *GRADE* emerged as a significant predictor, the limited impact of *NCAATOURN* and *PFIVE* underscores the importance of individual performance over general collegiate exposure in shaping signing bonus outcomes.

The second regression analyzed the impact of performance metrics on *DIFF*, focusing on *HR*, *OPS*, *BABIP*, and *SOBB*. This model's R-squared value was 0.0568, explaining 5.68% of the

variation in *DIFF*. While still modest, the inclusion of performance variables provided deeper insights into the factors influencing signing bonus deviations.

Among the performance metrics, *HR* emerged as a significant positive predictor of *DIFF*. The coefficient for *HR* was 0.0064 ($p = 0.011$), indicating that for each additional home run, *DIFF* increases by approximately 0.64%. For the average signing bonus in the dataset, which is \$1,239,598, this translates to an increase of \$7,934 per additional home run. This finding underscores the importance of power-hitting ability in draft negotiations. Players with higher *HR* totals are perceived as having greater offensive impact, making them more likely to secure bonuses above their slot values. Teams appear willing to invest in players who demonstrate strong power-hitting potential, reflecting the premium placed on this skill in professional baseball.

In contrast, other performance metrics did not reach statistical significance. The lack of significance for *OPS* may be attributed to its composite nature, which combines on-base percentage and slugging percentage. While *OPS* is a valuable metric for evaluating overall offensive performance, its overlap with other variables may dilute its independent explanatory power.

Similarly, *BABIP* and *SOBB* did not show significant relationships with *DIFF*. These results suggest that while these metrics capture specific aspects of player performance, such as batting effectiveness and strikeout-to-walk ratios, they may not directly translate to signing bonus deviations. The potential multicollinearity among these metrics could also limit their individual impact.

The findings from this model highlight the value of isolating specific performance metrics, such as *HR*, that directly influence team perceptions and negotiation leverage. The

modest R-squared value further emphasizes the need to incorporate additional contextual and demographic variables to improve the model's explanatory power.

The third regression combined both demographic and performance variables, including *GRADE*, *NCAATOURN*, *PFIVE*, *HR*, *OPS*, *BABIP*, and *SOBB*. This comprehensive model aimed to provide a holistic understanding of the factors influencing *DIFF*. The R-squared value for this model was 0.0665, explaining 6.65% of the variation in *DIFF*. While still limited, this model offered the most robust analysis of the dataset.

In this combined regression, *GRADE* and *HR* emerged as significant predictors. *GRADE* retained its negative relationship with *DIFF*, with a coefficient of -0.0232 ($p = 0.054$). This result reinforces the trend that players in higher grades receive bonuses closer to or below their slot values. The reduced leverage of older players, particularly seniors, likely reflects teams' preference for younger prospects with greater perceived growth potential.

HR also remained a significant positive predictor, with a coefficient of 0.0064 ($p = 0.011$). This finding aligns with the results of the performance-focused regression, emphasizing the critical role of power-hitting in draft negotiations. Players who demonstrate strong home run totals are more likely to command bonuses above their slot values, reflecting teams' confidence in their offensive potential.

Other variables, including *NCAATOURN*, *PFIVE*, *OPS*, *BABIP*, and *SOBB*, did not show statistically significant effects in the combined model. These results suggest that general collegiate exposure, whether through tournament participation or conference affiliation, does not independently drive significant deviations in signing bonuses. Similarly, the lack of significance for *OPS*, *BABIP*, and *SOBB* indicates that these metrics may overlap with other predictors, limiting their individual explanatory power.

The regression results provide several actionable insights for players and teams. For players, the findings highlight the importance of age, grade level, and performance metrics in shaping negotiation outcomes. Younger players, particularly those with strong power-hitting ability, are more likely to secure bonuses above their slot values. These players can leverage their youth and standout performance metrics, such as home runs, to strengthen their negotiation positions.

For teams, the analysis underscores the value of integrating performance metrics with contextual factors to refine draft strategies. Teams can prioritize players with demonstrated power-hitting ability while considering the trade-offs associated with age and grade level. Understanding the limited impact of general collegiate exposure, such as tournament participation or conference affiliation, can help teams focus on individual performance metrics that directly correlate with future potential.

Despite these insights, the analysis has limitations. The low R-squared values across all models indicate that a significant portion of the variation in *DIFF* remains unexplained. Unobserved factors, such as team-specific financial strategies, regional scouting practices, or negotiation dynamics, likely play a substantial role in determining signing bonuses. Additionally, potential multicollinearity among performance metrics may have diluted the effects of individual variables, complicating the interpretation of results.

Table 5: MLB Debut Outcomes by Signing Type

Signing Type	Number of Players	Percentage of Sample	MLB Debut Count	MLB Debut Percentage
UnderSlot	253	58.2%	129	50.0%
Over/AtSlot	182	41.8%	96	52.7%

Table 6: MLB Debut Predicted by Signing Bonus Difference

Probit regression

Number of obs = **435**LR chi2(1) = **0.04**Prob > chi2 = **0.8394**Log likelihood = **-295.34412**Pseudo R2 = **0.0001**

DEBUT	Coefficient	Std. err.	z	P> z 	[95% conf. interval]	
DIFF	-.0585535	.2891207	-0.20	0.840	-.6252196	.5081126
_cons	-.2157735	.063552	-3.40	0.001	-.3403331	-.0912139

Table 7: MLB Debut Predicted by Signing Bonus Difference and Home Runs

Probit regression

Number of obs = **435**LR chi2(2) = **0.04**Prob > chi2 = **0.9786**Log likelihood = **-295.34303**Pseudo R2 = **0.0001**

DEBUT	Coefficient	Std. err.	z	P> z 	[95% conf. interval]	
DIFF	-.0598532	.2904645	-0.21	0.837	-.629153	.5094467
HR	.0004977	.0106249	0.05	0.963	-.0203268	.0213222
_cons	-.2210495	.1293361	-1.71	0.087	-.4745436	.0324446

Now, I turn to the aspect of this thesis that examines whether *DIFF* is a meaningful predictor of MLB debut outcomes. This analysis explores whether paying college position players above their slot value translates into a tangible payoff for MLB teams by increasing the likelihood that these players reach the major leagues. By focusing on this relationship, I aim to determine if teams' financial investments in draft bonuses yield measurable returns in terms of player progression to the MLB.

Table 5 examines MLB debut outcomes based on signing type, comparing players who signed under slot value to those who signed at or above slot value. Across the dataset, 58.2% of players signed for under slot value, while 41.8% signed at or above slot value. Of those signing under slot value, 50.0% eventually made their MLB debut. In contrast, 52.7% of players signing at or above slot value reached the MLB. While this slight difference suggests a minor advantage for players who signed at or above slot value, the 2.7% gap indicates that signing type alone is not a strong predictor of MLB success.

To investigate whether *DIFF* predicts MLB debut outcomes, a Probit regression was conducted (Table 6). The results show that the coefficient for *DIFF* was -0.0586, indicating a slight negative relationship between signing bonus deviations and the likelihood of making an MLB debut. However, this relationship was not statistically significant ($p = 0.840$). The confidence interval (-0.6252 to 0.5081) highlights the uncertainty surrounding the effect size, while the pseudo R-squared value (0.0001) suggests that *DIFF* alone explains virtually none of the variation in MLB debut outcomes. These findings imply that signing bonus differences, as measured by *DIFF*, do not meaningfully influence whether a player reaches the MLB.

Building on this analysis, home runs (HR), previously identified as the only performance metric significantly correlated with *DIFF*, was added to a second Probit regression model to

assess its combined effect with DIFF on MLB debut outcomes (Table 7). In this model, the coefficient for DIFF remained negative (-0.0599) and statistically insignificant ($p = 0.837$). Similarly, the coefficient for HR was 0.0005, indicating a negligible positive relationship with MLB debut outcomes that also failed to reach statistical significance ($p = 0.963$). The pseudo R-squared value again remained extremely low (0.0001), emphasizing the limited explanatory power of this model. These results suggest that while DIFF and HR may be critical factors in draft negotiations, they do not directly predict whether a player reaches the MLB.

The results in Table 5 align with these regression findings. The near-equal debut rates between players signing under slot value (50.0%) and those signing at or above slot value (52.7%) reinforce the conclusion that signing bonuses alone are not reliable predictors of long-term success. Players who received higher bonuses relative to their draft slot did not demonstrate significantly better outcomes in terms of reaching the MLB. These results suggest that other variables, such as minor league performance, draft round, or organizational development strategies, play a more substantial role in determining MLB career trajectories.

This analysis highlights the complexity of predicting MLB success based on signing bonuses or isolated performance metrics. While DIFF provides insight into negotiation outcomes and HR correlates with higher signing bonuses, neither variable serves as a reliable predictor of MLB debut outcomes. The lack of significance in both regression models underscores a disconnect between the metrics valued during draft negotiations and those that determine long-term professional success.

6. Discussion

These results have several implications for stakeholders in the MLB draft process. Agents and players should consider leveraging performance metrics such as home runs and grade level during negotiations. Home runs, often associated with offensive potential, can drive higher bonuses, with each additional home run significantly correlating to a \$7,934 increase in signing bonus. Grade level also significantly impacts leverage, with younger players holding a clear advantage over older student-athletes, as each increase in grade level results in an average reduction of \$35,948 in signing bonus. Also, highlighting how a player's unique abilities that match team-specific needs should be a priority, with negotiations tailored to emphasize these strengths effectively.

For players, skill development is critical, particularly in areas like power-hitting, where metrics such as exit velocity and launch angle play a defining role in performance and draft appeal. As noted by Rapsodo, "The ideal power profile for most hitters includes an exit velocity of at least 90 mph, combined with a launch angle between 25 and 35 degrees. Statcast shows that hitters who consistently hit balls within that launch angle range are producing the highest slugging percentages." This data highlights the benefits of optimizing these metrics, as hitters meeting these thresholds demonstrate enhanced offensive productivity and increased value to teams. Players training to improve their exit velocity and fine-tune their launch angles can better align with MLB standards, making them more attractive prospects. By focusing on targeted drills to improve the factors of exit velocity and optimal launch angle, draft prospects can become more valuable during bonus negotiations and long-term player development.

For teams, understanding the factors that influence signing costs is critical. Metrics like home runs and grade are useful tools for negotiations but do not reliably predict MLB success, as

players signing above-slot bonuses exhibit only a 2.7% higher MLB debut rate compared to those signing below-slot. This highlights the need to identify additional value factors, such as leadership, work ethic, and mentality when evaluating prospects. Teams should also reevaluate the effectiveness of exceeding bonus pool caps, as paying over-slot bonuses does not guarantee better MLB outcomes and comes with penalties such as taxes and restrictions on future draft flexibility. Instead, resources might be better allocated to initiatives like international scouting, which uncovers undervalued talent at lower costs, or minor league development. Also, incorporating advanced metrics, such as Barrel Percentage (Barrel%) and Chase Rate (O-Swing%), into player evaluations can help identify hidden value in prospects. These advanced statistics are strong predictors of success and often align more closely with MLB performance than traditional metrics. By leveraging these insights, teams may uncover undervalued players who can be acquired for less than their potential suggests, maximizing the efficiency of their draft or roster strategies. Ultimately, players, agents, and teams should adopt data-driven, adaptive strategies in order to navigate the draft process optimally.

7. Conclusion

This thesis analyzed two key aspects of the MLB draft process: how NCAA Division I college position players maximize their signing bonuses and whether those bonuses predict future success, as measured by MLB debut likelihood. By examining college prospects drafted in the top five rounds between 2012 and 2022, this study provides valuable insights for players, agents, and teams. The findings reveal that signing bonuses are primarily shaped by performance metrics and leverage factors, particularly grade level and home runs. Each increase in grade, such as moving from sophomore to junior or junior to senior, decreases the signing bonus relative to the slot value by 2.93%, equivalent to \$35,948 for the dataset's average signing bonus of \$1.3 million dollars. This demonstrates that younger players have greater negotiation leverage due to their eligibility and perceived developmental potential, while seniors face reduced bargaining power. Additionally, each additional home run increases the signing bonus relative to slot value by 0.64%, equivalent to \$7,934. This reflects the premium placed on power-hitting ability during draft negotiations, as teams perceive players with higher home run totals as having greater immediate impact.

Despite the influence of these factors on signing bonuses, the study found that discrepancies between signing bonuses and slot values (*DIFF*) are not significant predictors of MLB success. An analysis of MLB debut outcomes revealed only a 2.7% difference in debut rates between players signing above slot value (52.7%) and those signing below (50.0%). Regression results confirmed that *DIFF* and home runs, while critical in financial negotiations, do not meaningfully influence whether a player reaches the majors. These findings suggest that other factors, such as minor league performance, positional versatility, and organizational development strategies, play a more substantial role in determining MLB career trajectories.

Future research should expand upon these findings by identifying additional factors that influence signing bonuses and predict MLB debut probability before the draft. One key area of exploration involves the impact of team draft history and resource allocation strategies on signing bonuses. Teams that aggressively allocate a significant portion of their bonus pools to early-round picks may struggle to offer competitive bonuses to later selections, potentially limiting their leverage and negotiation power. Conversely, teams adhering strictly to slot values may affect players' ability to negotiate for bonuses aligned with their talent, potentially leaving high-value players undercompensated. Research could evaluate whether teams with aggressive spending strategies or conservative adherence to slot values demonstrate better outcomes in terms of MLB debut rates for drafted players.

Another critical area involves the influence of MLB scouting grades (20-80 scale) on both signing bonuses and debut probabilities. Scouting grades, which evaluate a player's tools such as hitting, power, speed, defense, and arm strength, are widely used by teams to assess talent but remain underexplored as predictors of signing bonuses. For instance, a high scouting grade in power-hitting may correlate with above-slot bonuses, as teams view power as a premium tool. Similarly, understanding how lower grades in certain categories, such as speed or defense, might negatively impact bonus negotiations could offer insights into how teams weigh different tools during evaluations. Future research could explore whether these grades align with players' actual potential to succeed at the MLB level, providing teams with a more reliable framework for decision-making.

By examining these factors, future studies could enhance understanding of the MLB draft process and provide actionable insights. This research would allow teams to optimize resource allocation, refine player evaluations, and create more effective strategies for maximizing the

return on their draft investments. It would also enable players and agents to better understand how team strategies and evaluations shape bonus negotiations, empowering them to advocate for fair compensation.

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